UNIVERSIDAD
NACIONAL
DE COLOMBIA

Laboratory 1

Structural Design

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Large-Scale Software Architecture

2026-I

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1. Objective

To construct a **Component-and-Connector (C&C) view** of a real-world large-scale System of Systems (SoS), developing the ability to identify and represent systems, components, connectors, and their interactions across multiple architectural tiers.

This laboratory uses the **European Rail Traffic Management System (ERTMS)** as a guided case study. The goal is not only to reproduce a known structure, but to reason about it: to understand why each element exists, what role it plays, and how it relates to the others. And then to represent that understanding through a precise C&C view.

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2. Background

2.1 The ERTMS as a System of Systems

The **European Rail Traffic Management System (ERTMS)** is one of the most widely cited examples of a large-scale, federated System of Systems. It was designed to enable interoperable rail traffic management across different European countries and operators, integrating dozens of previously incompatible national systems.

In SoS terms, ERTMS is composed of multiple **constituent systems** — each operationally and managerially autonomous — that collaborate to achieve a collective mission: safe and efficient cross-border rail traffic coordination. Each constituent system can operate independently, but contributes specific capabilities to the global SoS behavior through well-defined information exchanges.

2.2. The C&C View

A **Component-and-Connector (C&C) view** is an architectural representation that describes the runtime structure of a system by identifying:

- **Components:** The active elements that perform computation, sensing, control, or coordination. In a cyber-physical SoS like ERTMS, components may be software (services, modules), hardware (onboard units, gateways), or physical (sensors, actuators, track equipment).
- **Connectors:** The interaction mechanisms through which components exchange data or control information. Connectors describe *what* is exchanged and *between which elements*, abstracting away implementation details such as protocols or middleware.

The C&C view does not represent deployment infrastructure (servers, networks, cloud instances) — those belong to the Deployment View. It focuses exclusively on elements with an active role in computation, sensing, or control.

2.3. 5-Tier Structure of the ERTMS

The ERTMS architecture can be analyzed through a **5-tier structure** that organizes its components by their operational role:

Tier	Name	Role
T_1	Presentation	User-facing interfaces for passengers, operators, and drivers
T_2	Communication	Inter-system and inter-component interaction infrastructure
T_3	Logic	Business logic, control services, and domain services
T_4	Data	Persistent storage, databases, and data lakes
T_5	Physical	Onboard units, sensors, actuators, and track-side equipment

This tiered structure is not a deployment model — it is an organizational lens for the C&C view, grouping components by their architectural role within the SoS.

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3. Reference Elements

The following is a **partial reference** of components identified in the ERTMS architecture. These elements are a starting point — they do not represent the complete structure. Part of this laboratory's challenge is to reason about what is missing and complete the C&C view accordingly.

T_1 — Presentation

- Portal *(passenger-facing web interface)*
- Dashboard *(operator monitoring interface)*
- Driver Interface *(onboard display for train drivers)*

T_2 — Communication

- API Gateway *(central entry point routing requests from presentation components to logic services)*

T_3 — Logic

- passengers-ms, routes-ms, trains-ms *(core domain microservices)*
- position-time-ms, tickets-ms *(operational microservices)*
- MAS — Movement Authority Service *(authorizes train movement along a given corridor)*

T_4 — Data

- passengers-db, routes-db, trains-db
- position-time-db
- DL — Data Lake *(aggregated operational and historical data)*

T_5 — Physical

- Onboard Unit *(main computing unit aboard the train)*
- Train Sensor *(detects speed, position, and physical conditions)*
- Brake Actuator *(receives control commands and acts on the physical train)*
- Balise *(track-side transponder that communicates position reference data to passing trains)*

> **Note:** These elements are representative, not exhaustive. The actual ERTMS architecture includes additional components at each tier. As part of this laboratory, students are expected to identify and incorporate additional components and connectors that they consider architecturally relevant and justified.

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4. Activities

4.1. Activity 1 — Structural Analysis

Before constructing any representation, analyze the reference elements provided above and discuss the following questions as a group:

1. Which of the listed elements are **software components**, which are **hardware components**, and which are **physical components**? Justify each classification.
2. For each tier, identify at least **two connectors** that you would expect to exist between components. What information does each connector carry? In which direction does it flow?
3. Which components interact **within the same tier** (intra-tier connectors), and which require **cross-tier connectors**? What does this imply architecturally?
4. Are there components in the reference list that you believe are **incomplete or missing**? Which additional components would you expect to find at each tier, and why?

4.2. Activity 2 — C&C View Construction

Construct the **C&C view of the ERTMS** as a System of Systems. This view must represent the full architecture across the five tiers, incorporating both the reference elements and any additional components your team identifies as necessary.

The C&C view must:

- Clearly delimit the **five tiers** (T_1 through T_5) as distinct architectural layers.
- Represent all **components** with their names and type (software / hardware / physical).
- Represent all **connectors** between components, indicating:
  - The **direction** of information flow.

    - The **nature** of the connector: data flow, control command, dependency, or event notification.
- Include the **additional components and connectors** identified by your team, beyond the reference elements provided.
- Be internally consistent: every connector must have a clearly identified source component and a target component.

4.3. Activity 3 — Reflection

Once the C&C view is complete, discuss and document brief answers to the following:

1. Which tier concentrates the highest **architectural complexity** in terms of number of components and connectors? What does this reveal about the system's design priorities?
2. Where in the C&C view would you expect **availability** to be most critical? How does the architecture address (or fail to address) this?
3. How does the ERTMS structure, as a SoS, differ from a conventional monolithic or microservices architecture? What makes it a SoS and not simply a large distributed system?

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5. Delivery

A **C&C View** for the *ERTMS SoS*.

5.1. Submission Format

The view must be submitted via Google Classroom. File in *.pdf* format named **lab1-team-X**, where X = team (1a, 1b, ... 2a, 2b, ...).

5.2. Delivery Deadline

Friday, March 27, 2026, before 23h59.